

keep their hands free and they worked on the same technology that went into telephone receivers.

While we might broadly use the classifications mentioned above to differentiate between different types of headphones, there are finer details within the two broad categories that appeal to specific users.

Circumaural headphones have ear pads that are circular, ellipsoid or even squarish in their shape which surround the ear. Due to this design, it is possible to make circumaural headphones that can block outside noise completely without hampering sound quality. That being said, these tend to be quite heavy at times and are mostly used by musicians.

Supra-aural headphones are slightly smaller as their pads press on the ear, rather than going around it. But due to this design, these earphones tend to be painful if worn for longer durations and can mainly be improved depending on the cushioning material being used. Headphones is a vast and interesting topic and we'll get into more depth in Chapter 3.



Creative Soundblaster Roar Bluetooth speakers

Speakers

There is almost no difference in the technology that goes behind large loudspeakers you see in concerts and recording studios, compared to normal sized home speakers, USB speakers and the like. What differentiates them is the design principles followed for each, and the component choices

made to adhere to that principle. For example, a USB powered speaker has to keep in mind the maximum power capacity that its USB connection will allow, and choose, or build, equipment accordingly that will consume less power or have a battery pack in-built. Similarly, bluetooth speakers have the goal of maximising duration of playback along with audio quality. A more detailed look into speakers has been taken in the next chapter.

Amplifier

Powered amps are electronic amplifiers that amplify low power electronic audio signals. These initial inputs usually have a frequency range of 20Hz to 20,000Hz, which is amplified in power by the amplifier unit to be strong enough to drive loudspeakers and send the audio across to listeners.